

# 1 Two stochastic epidemic models

We consider two discrete-time, branching-process models for the spread of an infectious disease. Time is measured in days. Let  $I_t \geq 0$  denote the number of new infections (incidence) on day  $t$ , and let  $R_t > 0$  be the effective reproduction number on day  $t$ .

The *serial interval weights*  $w_s > 0$  ( $s = 1, 2, \dots$ ) satisfy  $\sum_{s=1}^{\infty} w_s = 1$  and encode the probability that a secondary infection appears  $s$  days after the primary case. Write

$$F_r = \sum_{s=1}^r w_s$$

for the cumulative serial interval distribution at lag  $r$ , so  $1 - F_r = \sum_{s=r+1}^{\infty} w_s$  is the probability that transmission occurs after day  $r$ .

The *dispersion parameter*  $k > 0$  controls individual-level variation in infectiousness. Small  $k$  corresponds to high heterogeneity (superspreading); as  $k \rightarrow \infty$  the negative-binomial offspring distribution converges to a Poisson.

**Remark.** Throughout,  $\text{NB}(\text{mean} = \mu, \text{disp} = k)$  denotes the negative-binomial distribution with probability mass function

$$\mathbb{P}(X = n) = \binom{n+k-1}{n} \left( \frac{k}{k+\mu} \right)^k \left( \frac{\mu}{k+\mu} \right)^n, \quad n = 0, 1, 2, \dots,$$

mean  $\mu$ , variance  $\mu + \mu^2/k$ , and probability generating function (pgf)  $G(s) = (k / (k + \mu(1-s)))^k$ . Similarly,  $\text{Gamma}(\alpha, \beta)$  denotes the Gamma distribution with shape  $\alpha$  and rate  $\beta$ , so density  $f(x) = \beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ , mean  $\alpha/\beta$ , and variance  $\alpha/\beta^2$ .

## 2 SSE model (Superspreading Events)

In the *superspreading-events* (SSE) model, the force of infection on day  $t$  is the weighted sum of past incidence,

$$\lambda_t = \sum_{s=1}^{\infty} w_s I_{t-s}, \tag{1}$$

and new infections are negative-binomially distributed:

$$I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t). \tag{2}$$

The conditional mean of  $I_t$  is  $R_t \lambda_t$  and its variance is  $R_t \lambda_t + R_t^2 \lambda_t / k$ , so the NB inflates the Poisson variance by the factor  $1 + R_t \lambda_t / k$ , capturing superspreading at the population level.

## 3 SSI model (Superspreading Individuals)

In the *superspreading individuals* (SSI) model, heterogeneity enters at the individual level. Each infected individual has a latent infectivity that is Gamma distributed with mean 1 and variance  $1/k$ .

Each cohort of  $I_t$  infected individuals then has an aggregate infectivity

$$Y_t \mid I_t \sim \text{Gamma}(\text{shape} = k I_t, \text{rate} = k), \tag{3}$$

so  $\mathbb{E}[Y_t \mid I_t] = I_t$  and  $\text{Var}(Y_t \mid I_t) = I_t / k$ . New infections arise as a Poisson process driven by past infectivity,

$$I_t \mid (Y_s)_{s < t} \sim \text{Poisson} \left( R_t \sum_{s=1}^{\infty} w_s Y_{t-s} \right). \tag{4}$$

**Remark.** Marginalising  $Y_t$  from the SSI model recovers a negative-binomial offspring distribution (by the Poisson–Gamma mixture identity; see below), so the two models share the same mean–variance structure at the population level while differing in the *mechanism* generating overdispersion.

## 4 Extinction probability

Henceforth assume the reproduction number is constant,  $R_t = R_0$ . We study the probability that the epidemic goes extinct, starting from a single infectious case on day 0.

By branching-process theory, the extinction probability  $q$  is the smallest non-negative fixed point of the pgf of the offspring distribution. For both models the offspring distribution is NB(mean =  $R_0$ , disp =  $k$ ), whose pgf is  $G(s) = (k/(k + R_0(1 - s)))^k$ . Setting  $q = G(q)$  gives

$$q = \left(1 + \frac{R_0}{k}(1 - q)\right)^{-k}. \quad (5)$$

If  $R_0 \leq 1$  the unique solution is  $q = 1$  (extinction is certain); if  $R_0 > 1$  there is a unique solution  $q \in (0, 1)$ , so the probability of a major outbreak is  $1 - q > 0$ .

### 4.1 Extinction probability after observing incidence on days 0 to $r$

Suppose we observe incidence  $I_0, I_1, \dots, I_r$  on the first  $r + 1$  days, with  $I_0 \geq 1$ . We compute the updated extinction probability  $q_r$  for the residual epidemic — the probability that the chain triggered by the observed cases eventually dies out.

A standard branching-process argument applies in both models: each remaining offspring of the observed cases, once born, initiates an independent fresh epidemic governed by the original NB( $R_0$ ,  $k$ ) offspring distribution and goes extinct with the standard probability  $q$  (the smallest fixed point of  $G$ ). The conditioning event constrains only the offspring of the observed cases on days  $0, \dots, r$ ; it does not propagate to later generations. Therefore

$$q_r = G_r(q),$$

where  $G_r$  is the pgf of the (conditional) total remaining offspring of the observed cases. The two models differ in the form of  $G_r$ .

#### 4.1.1 SSE model

In the SSE model the offspring process decomposes into independent contributions: each of the  $I_s$  individuals infected on day  $s$  generates a number of offspring at lag  $\ell$  that is NB(mean =  $R_0 w_\ell$ , disp =  $k w_\ell$ ), independently across individuals, donor days, and lags. By the NB closure property under a common success probability  $p = k w_\ell / (k w_\ell + R_0 w_\ell) = k / (k + R_0)$  (independent of  $\ell$ ), the population-level NB( $R_0 \lambda_t$ ,  $k \lambda_t$ ) law of  $I_t$  is recovered by summing all per-individual-per-lag contributions.

Conditioning on the observed counts  $I_0, \dots, I_r$  pins down only those contributions  $(s, \ell)$  with  $s + \ell \leq r$ . Per-individual-per-lag contributions with  $s \in \{0, \dots, r\}$  and  $\ell > r - s$  are independent of the observations and retain their original distributions. The total remaining offspring of the observed cases is therefore

$$\sum_{s=0}^r \sum_{i=1}^{I_s} \sum_{\ell=r-s+1}^{\infty} \text{NB}(\text{mean} = R_0 w_\ell, \text{disp} = k w_\ell) = \text{NB}(\text{mean} = R_0 \Lambda, \text{disp} = k \Lambda), \quad (6)$$

where the pooled effective weight is

$$\Lambda = \sum_{s=0}^r I_s (1 - F_{r-s}). \quad (7)$$

With the convention  $F_0 = 0$ , the  $s = r$  term enters with full weight, since none of its offspring have had a chance to appear yet.

The pgf is  $G_r(s) = (1 + (R_0/k)(1 - s))^{-k\Lambda} = G(s)^\Lambda$ , so

$$q_r = G_r(q) = q^\Lambda = q^{\sum_{s=0}^r I_s (1 - F_{r-s})}. \quad (8)$$

The earlier special case (a single case on day 0 followed by  $r$  days without cases) corresponds to  $I_0 = 1$ ,  $I_1 = \dots = I_r = 0$ , giving  $\Lambda = 1 - F_r$  and recovering  $q_r = q^{1 - F_r}$ . In the same direction, as the zero streak extends ( $r \rightarrow \infty$ ) the contributions  $I_s (1 - F_{r-s})$  all vanish and  $q_r \rightarrow 1$ : extinction approaches certainty.

#### 4.1.2 SSI model

For the SSI model the joint posterior of  $(Y_0, \dots, Y_{r-1})$  is in general not available in closed form: as soon as  $I_d > 0$  on at least one day  $d \in \{1, \dots, r\}$ , the day- $d$  rate  $R_0 \sum_{s=1}^{d-1} w_{d-s} Y_s$  couples the latents. We therefore present an MCMC estimator of  $q_r$  here as the canonical SSI workflow. For the special cases of cases on day 0 only, on day 0 and one later day, and on day 0 and two later days, closed forms are however available; these are deferred to § 6 below.

**MCMC posterior.** We sample  $(Y_0, \dots, Y_{r-1})$  from their joint posterior given  $(I_0, \dots, I_r)$  using a Hamiltonian Monte Carlo sampler (implemented in PyMC). Note that  $Y_r$ , the aggregate infectivity of the day- $r$  cohort, has not yet contributed to any observed outcome; its posterior equals its prior,

$$Y_r \sim \text{Gamma}(kI_r, k),$$

and is sampled independently.

**Remaining offspring from observed cases.** Given  $(Y_s)_{s=0}^r$ , the future infections generated by the observed cases arrive as independent Poisson streams: cohort  $s$  contributes at lag  $\ell \geq r - s + 1$  at rate  $R_0 w_\ell Y_s$ . Summing over cohorts and lags gives total remaining offspring

$$I_{\text{rem}} \mid (Y_s)_{s=0}^r \sim \text{Poisson}(R_0 \Lambda), \quad \Lambda = \sum_{s=0}^r Y_s (1 - F_{r-s}), \quad (9)$$

exactly the same pooled-weight formula as in the SSE model.

**Extinction probability and MCMC estimator.** Each remaining offspring initiates an independent fresh epidemic that goes extinct with the standard probability  $q$ . Hence the conditional extinction probability of the residual epidemic, given the latent infectivities, is the pgf of  $\text{Poisson}(R_0 \Lambda)$  evaluated at  $q$ :

$$q_r \mid (Y_s)_{s=0}^r = e^{R_0 \Lambda (q-1)}. \quad (10)$$

Marginalising over the posterior of  $(Y_s)$  and estimating by a Monte Carlo average over  $M$  MCMC draws,

$$\hat{q}_r = \frac{1}{M} \sum_{m=1}^M e^{R_0 \Lambda^{(m)} (q-1)}, \quad \widehat{\text{PMO}} = 1 - \hat{q}_r, \quad (11)$$

where  $\Lambda^{(m)} = \sum_{s=0}^r Y_s^{(m)} (1 - F_{r-s})$ .

**Remark** (Consistency with the day-0-only analytic formula). When  $I_1 = \dots = I_r = 0$  the observations force  $Y_1 = \dots = Y_r = 0$ , so  $\Lambda = Y_0(1 - F_r)$  and the posterior is  $Y_0 \sim \text{Gamma}(kI_0, k + R_0 F_r)$ . Taking the expectation of  $e^{R_0 Y_0 (1 - F_r)(q-1)}$  with respect to this posterior via the Gamma moment-generating function gives

$$q_r = \left(1 - \frac{R_0(1 - F_r)(q-1)}{k + R_0 F_r}\right)^{-kI_0} = \left(1 + \frac{R_0(1 - F_r)}{k + R_0 F_r}(1 - q)\right)^{-kI_0},$$

which matches the closed-form result (20).

## 5 Probability of major outbreak under model uncertainty

So far we have computed the probability of a major outbreak conditional on a chosen model (SSE or SSI). In practice the underlying mechanism generating overdispersion is unknown, and we may wish to combine the two via Bayesian model averaging. Let  $\pi_{\text{SSE}}, \pi_{\text{SSI}} \in [0, 1]$  with  $\pi_{\text{SSE}} + \pi_{\text{SSI}} = 1$  be prior probabilities on the two models. Given the observed history  $(I_0, \dots, I_r)$ , Bayes' theorem yields posterior model probabilities

$$\pi_M^* = \frac{\pi_M L_M}{\pi_{\text{SSE}} L_{\text{SSE}} + \pi_{\text{SSI}} L_{\text{SSI}}}, \quad L_M = \mathbb{P}_M(I_1, \dots, I_r \mid I_0), \quad M \in \{\text{SSE}, \text{SSI}\}. \quad (12)$$

The model-averaged probability of a major outbreak is then

$$\text{PMO}_{\text{avg}} = \pi_{\text{SSE}}^* \cdot \text{PMO}_{\text{SSE}} + \pi_{\text{SSI}}^* \cdot \text{PMO}_{\text{SSI}}, \quad (13)$$

where  $\text{PMO}_M = 1 - q_r^M$  as derived above.

## 5.1 Likelihood of the observed history

**SSE model.** Conditional on past incidence, the day- $t$  count satisfies  $I_t \mid I_{<t} \sim \text{NB}(\text{mean} = R_0 \lambda_t, \text{disp} = k \lambda_t)$  with  $\lambda_t = \sum_{s=1}^{t-1} w_{t-s} I_s$ , so the joint likelihood factorises into a product of negative-binomial pmfs,

$$L_{\text{SSE}}(I_1, \dots, I_r \mid I_0) = \prod_{t=1}^r \text{NB}(I_t; R_0 \lambda_t, k \lambda_t). \quad (14)$$

This closed form is available for any history.

**SSI model.** The likelihood involves the latent infectivities  $(Y_s)_{s=0}^{r-1}$  and is in general not available in closed form. For genuinely general histories, an importance-sampling estimate using draws from the posterior  $p(Y_{0:r-1} \mid I_{0:r})$  produced by the MCMC sampler of the previous section is feasible but is not implemented here. Closed forms are however available in the same three special cases as for  $q_r$  (cases on day 0 only, on day 0 and one later day, on day 0 and two later days); these are derived in § 6 below.

## 5.2 Simulation-based estimator

For general histories, where the SSI likelihood is unavailable in closed form, we use a rejection-sampling estimator that simultaneously realises the posterior model weighting. Repeat the following in batches of size  $b$ :

1. Draw the per-batch model split as a Binomial:  $b_{\text{SSE}} \sim \text{Bin}(b, \pi_{\text{SSE}})$  and  $b_{\text{SSI}} = b - b_{\text{SSE}}$ .
2. Run  $b_{\text{SSE}}$  vectorised SSE forward simulations from  $t = 0$  seeded with  $I_0$  index cases, retaining only those whose simulated incidence on days  $1, \dots, r$  exactly matches the observed  $I_1, \dots, I_r$ . Likewise for  $b_{\text{SSI}}$  SSI simulations.
3. For each retained simulation, continue forward until either the single-step incidence reaches the major-outbreak threshold (resolved as “major”) or the most recent  $L = \text{len}(w)$  steps are all zero (resolved as “extinct”).

Continue accumulating batches until a target number of resolved-and-matching simulations is reached. The acceptance rate of model  $M$ ’s simulations is  $L_M$  in expectation; with a Binomial split by prior, the expected number of accepted  $M$ -sims is proportional to  $\pi_M L_M$ , so the composition of the accepted pool matches the posterior  $\pi_M^*$ . The pooled estimator

$$\widehat{\text{PMO}}_{\text{avg}} = \frac{N_{\text{SSE}}^{\text{maj}} + N_{\text{SSI}}^{\text{maj}}}{N_{\text{SSE}}^{\text{acc}} + N_{\text{SSI}}^{\text{acc}}} = \hat{\pi}_{\text{SSE}}^* \widehat{\text{PMO}}_{\text{SSE}} + \hat{\pi}_{\text{SSI}}^* \widehat{\text{PMO}}_{\text{SSI}} \quad (15)$$

follows immediately, where  $\hat{\pi}_M^* = N_M^{\text{acc}} / (N_{\text{SSE}}^{\text{acc}} + N_{\text{SSI}}^{\text{acc}})$  is the implied posterior estimate and  $\widehat{\text{PMO}}_M = N_M^{\text{maj}} / N_M^{\text{acc}}$  is the per-model PMO from the accepted  $M$ -sims.

## 6 SSI model: closed forms for histories with cases on a few days

For the SSI model, when non-zero incidence is observed on day 0 and on a small number of later days, the joint posterior of the latent infectivities still factorises (or factorises as a finite mixture) and both the model evidence  $L_{\text{SSI}}$  and the residual extinction probability  $q_r$  admit closed forms. This section treats three such cases in turn: case (i) (§ 6.1), incidence on day 0 only; case (ii) (§ 6.2), incidence on day 0 and one later day; and case (iii) (§ 6.3), incidence on day 0 and two later days.

**Strategy.** Each derivation follows the same five steps:

1. *Conditional likelihood given the latents.* Multiply the per-day Poisson pmfs to obtain  $\mathbb{P}(\text{obs} \mid Y_0, Y_{i_1}, \dots, Y_{i_n})$ .
2. *Joint density of latents and observation.* Multiply by the independent Gamma priors  $Y_s \sim \text{Gamma}(kI_s, k)$  to get  $p(Y_0, Y_{i_1}, \dots, Y_{i_n}, \text{obs})$ , with all normalising constants visible.
3. *Likelihood (model evidence).* Integrate the joint density over the latents using the Gamma normalisation  $\int_0^\infty Y^{\alpha-1} e^{-\beta Y} dY = \Gamma(\alpha)/\beta^\alpha$  in each variable.
4. *Posterior of the latents.* Divide the joint density by the likelihood to read off the posterior; in the cases below this yields independent Gammas (or a finite mixture of independent-Gamma products in case (iii)).
5. *Residual extinction probability and PMO.* The conditional extinction probability given the latents is  $q_r \mid (Y_s) = e^{-R_0(1-q)\Lambda}$  with  $\Lambda = \sum_{s=0}^r Y_s(1-F_{r-s})$  (cf. the SSI MCMC derivation in § SSI model above); marginalising over the posterior via the Gamma MGF  $\mathbb{E}[e^{-tX}] = (1+t/\beta)^{-\alpha}$  for  $X \sim \text{Gamma}(\alpha, \beta)$  gives a closed-form  $q_r$ , and the probability of a major outbreak is  $\text{PMO} = 1 - q_r$ .

A key identity used throughout is that  $I_s = 0$  on day  $s$  forces  $Y_s = 0$  (since  $Y_s \mid I_s \sim \text{Gamma}(kI_s, k)$  collapses to a point mass at 0), so among the latents only those tied to non-zero observed days are non-degenerate. Likewise, the SSE likelihood (14) is closed-form for any history, so each subsection below ends with the model-averaged PMO via Bayes' theorem from the per-model likelihoods and PMOs.

## 6.1 Case (i): $I_0 \geq 1, I_1 = \dots = I_r = 0$

Only  $Y_0$  is non-degenerate; we abbreviate  $y = Y_0$ .

**Step 1: conditional likelihood.** Given  $y$ ,  $I_d$  is independently  $\text{Poisson}(R_0 w_d y)$  for  $d = 1, \dots, r$ , so

$$\mathbb{P}(I_1 = \dots = I_r = 0 \mid y) = \prod_{d=1}^r e^{-R_0 w_d y} = e^{-R_0 F_r y}. \quad (16)$$

**Step 2: joint density.** Multiplying (16) by the prior  $y \sim \text{Gamma}(kI_0, k)$ ,

$$p(y, \text{obs}) = \frac{k^{kI_0}}{\Gamma(kI_0)} y^{kI_0-1} e^{-(k+R_0 F_r)y}. \quad (17)$$

**Step 3: likelihood.** Integrating  $y$  out of (17) via the Gamma normalisation,

$$L_{\text{SSI}}^{(i)} = \frac{k^{kI_0}}{\Gamma(kI_0)} \cdot \frac{\Gamma(kI_0)}{(k + R_0 F_r)^{kI_0}} = \left(1 + \frac{R_0 F_r}{k}\right)^{-kI_0}. \quad (18)$$

**Step 4: posterior.** Dividing (17) by (18) every constant cancels except the Gamma rate and shape, giving

$$y \mid \text{obs} \sim \text{Gamma}(kI_0, k + R_0 F_r). \quad (19)$$

The posterior rate  $k + R_0 F_r > k$  reflects the fact that observing no early transmission is evidence that  $y$  is small: the index cohort is likely a low-infectivity one.

**Step 5: residual extinction probability and PMO.** The conditional remaining-offspring rate is  $R_0 \Lambda = R_0(1 - F_r)y$ , so the conditional extinction probability is  $q_r \mid y = e^{-R_0(1-q)(1-F_r)y}$ . Marginalising over the posterior (19) via the Gamma MGF,

$$q_r = \mathbb{E}\left[e^{-R_0(1-q)(1-F_r)y}\right] = \left(1 + \frac{R_0(1-q)(1-F_r)}{k + R_0 F_r}\right)^{-kI_0}, \quad (20)$$

and  $\text{PMO} = 1 - q_r$ .

**Closed-form SSE counterpart.** Under the SSE model,  $I_t \sim \text{NB}(R_0 I_0 w_t, k I_0 w_t)$  independently across  $t = 1, \dots, r$  when  $I_1 = \dots = I_r = 0$ , so the general likelihood (14) reduces to

$$L_{\text{SSE}} = \prod_{t=1}^r \left( \frac{k}{k + R_0} \right)^{k I_0 w_t} = \left( 1 + \frac{R_0}{k} \right)^{-k I_0 F_r}. \quad (21)$$

The two likelihoods share the same functional shape but with different exponents: the SSE exponent  $k I_0 F_r$  is reduced by  $F_r \leq 1$  relative to the SSI exponent  $k I_0$ , reflecting the fact that the SSE total over days  $1, \dots, r$  has dispersion  $k I_0 F_r$  whereas the SSI total has dispersion  $k I_0$ . Combined with the SSE PMO  $1 - q^{I_0(1-F_r)}$  from § SSE model and the SSI PMO (20), the model-averaged PMO is

$$\text{PMO}_{\text{avg}} = \pi_{\text{SSE}}^* (1 - q^{I_0(1-F_r)}) + \pi_{\text{SSI}}^* \left[ 1 - \left( 1 + \frac{R_0(1-F_r)(1-q)}{k + R_0 F_r} \right)^{-k I_0} \right]. \quad (22)$$

When  $r = 0$ , both likelihoods reduce to 1, the posterior reduces to the prior, and the model-averaged PMO is just the prior-weighted average of the two per-model PMOs.

**Remark** (Consistency with the MCMC formula). When  $I_1 = \dots = I_r = 0$ , the observations force  $Y_1 = \dots = Y_r = 0$ , so  $\Lambda = Y_0(1 - F_r)$  and the general MCMC posterior of § SSI model collapses to (19). Taking the expectation of  $e^{-R_0(1-q)(1-F_r)Y_0}$  with respect to that posterior via the Gamma MGF recovers (20).

## 6.2 Case (ii): $I_0, I_i \geq 1$ for one later day $i$ , all others zero

We now extend to  $I_0 \geq 1$ ,  $I_i \geq 1$  for some  $i \in \{1, \dots, r\}$ , and  $I_d = 0$  for  $d \in \{1, \dots, r\} \setminus \{i\}$ . Only  $Y_0$  and  $Y_i$  are non-degenerate.

**Step 1: conditional likelihood.** Given  $(Y_0, Y_i)$ , the day- $d$  Poisson rate is  $R_0 w_d Y_0$  for  $d \in \{1, \dots, i\}$  and  $R_0(w_d Y_0 + w_{d-i} Y_i)$  for  $d \in \{i+1, \dots, r\}$ . Multiplying the day- $d$  Poisson pmfs,

$$\mathbb{P}(\text{obs} \mid Y_0, Y_i) = \underbrace{\prod_{d=1}^{i-1} e^{-R_0 w_d Y_0}}_{= e^{-R_0 F_{i-1} Y_0}} \cdot \underbrace{\frac{(R_0 w_i Y_0)^{I_i}}{I_i!} e^{-R_0 w_i Y_0}}_{\text{day-}i \text{ pmf}} \cdot \underbrace{\prod_{d=i+1}^r e^{-R_0(w_d Y_0 + w_{d-i} Y_i)}}_{= e^{-R_0(F_r - F_i)Y_0} e^{-R_0 F_{r-i} Y_i}}.$$

The  $Y_0$  exponentials collapse via  $F_{i-1} + w_i + (F_r - F_i) = F_r$ , so

$$\mathbb{P}(\text{obs} \mid Y_0, Y_i) = \frac{(R_0 w_i Y_0)^{I_i}}{I_i!} e^{-R_0 F_r Y_0} e^{-R_0 F_{r-i} Y_i}. \quad (23)$$

**Step 2: joint density.** Multiplying (23) by the priors  $Y_0 \sim \text{Gamma}(k I_0, k)$  and  $Y_i \sim \text{Gamma}(k I_i, k)$  (independent),

$$\begin{aligned} p(Y_0, Y_i, \text{obs}) &= \underbrace{\frac{(R_0 w_i)^{I_i}}{I_i!} \cdot \frac{k^{k I_0 + k I_i}}{\Gamma(k I_0) \Gamma(k I_i)}}_{\text{constants}} \\ &\times \underbrace{Y_0^{k I_0 + I_i - 1} e^{-(k + R_0 F_r) Y_0}}_{\text{Gamma kernel in } Y_0} \cdot \underbrace{Y_i^{k I_i - 1} e^{-(k + R_0 F_{r-i}) Y_i}}_{\text{Gamma kernel in } Y_i}. \end{aligned} \quad (24)$$

**Step 3: likelihood.** Integrating each Gamma kernel of (24) via the Gamma normalisation gives

$$L_{\text{SSI}}^{(\text{ii})} = \frac{(R_0 w_i)^{I_i}}{I_i!} \frac{\Gamma(k I_0 + I_i)}{\Gamma(k I_0)} \frac{k^{k I_0}}{(k + R_0 F_r)^{k I_0 + I_i}} \left( \frac{k}{k + R_0 F_{r-i}} \right)^{k I_i}. \quad (25)$$

Setting  $I_i = 0$  collapses the  $\Gamma$ -ratio and the second parenthesis to 1, recovering (18).

**Step 4: posterior.** Dividing (24) by (25), every constant cancels and the two Gamma kernels normalise independently:

$$Y_0 \mid \text{obs} \sim \text{Gamma}(kI_0 + I_i, k + R_0F_r), \quad Y_i \mid \text{obs} \sim \text{Gamma}(kI_i, k + R_0F_{r-i}). \quad (26)$$

The shape of  $Y_0$  is enriched by  $I_i$ : each observed offspring of the index cohort acts as a Bayesian data point tilting  $Y_0$  toward higher values. The  $Y_i$  posterior is the day-0-only update (19) applied to the day- $i$  cohort over its residual horizon  $r - i$ ; setting  $i = r$  gives  $F_{r-i} = F_0 = 0$ , so  $Y_r$  collapses to its prior, since the day- $r$  cohort has not had a chance to contribute.

**Step 5: residual extinction probability and PMO.** The remaining-offspring rate is  $R_0\Lambda$  with  $\Lambda = Y_0(1 - F_r) + Y_i(1 - F_{r-i})$ , so the conditional extinction probability is  $q_r \mid (Y_0, Y_i) = e^{-R_0(1-q)\Lambda}$ . Marginalising over the two independent Gamma factors of (26) via the Gamma MGF gives

$$q_r = \left(1 + \frac{R_0(1 - F_r)(1 - q)}{k + R_0F_r}\right)^{-(kI_0 + I_i)} \left(1 + \frac{R_0(1 - F_{r-i})(1 - q)}{k + R_0F_{r-i}}\right)^{-kI_i}, \quad (27)$$

and  $\text{PMO} = 1 - q_r$ .

**Remark** (Reduction to case (i)). Setting  $I_i = 0$  in (27) collapses the second factor to 1 and the first factor's exponent to  $-kI_0$ , recovering (20); likewise (25) reduces to (18). As the observation horizon extends ( $r \rightarrow \infty$  with  $i$  fixed), both factors of (27) tend to 1 and  $q_r \rightarrow 1$ .

### 6.3 Case (iii): $I_0, I_i, I_j \geq 1$ for two later days $i < j$ , all others zero

Finally, take  $I_0 \geq 1, I_i \geq 1, I_j \geq 1$  for some  $1 \leq i < j \leq r$ , and  $I_d = 0$  for  $d \in \{1, \dots, r\} \setminus \{i, j\}$ . Now only  $Y_0, Y_i, Y_j$  are non-degenerate.

**Step 1: conditional likelihood.** The conditional likelihood acquires an extra factor for the day- $j$  Poisson pmf, whose rate  $R_0(w_jY_0 + w_{j-i}Y_i)$  couples  $Y_0$  and  $Y_i$ . Repeating the telescoping argument of case (ii) for each latent separately,

$$\mathbb{P}(\text{obs} \mid Y_0, Y_i, Y_j) = \frac{(R_0w_iY_0)^{I_i}}{I_i!} \frac{(R_0(w_jY_0 + w_{j-i}Y_i))^{I_j}}{I_j!} e^{-R_0F_rY_0} e^{-R_0F_{r-i}Y_i} e^{-R_0F_{r-j}Y_j}. \quad (28)$$

Expanding the day- $j$  rate via the binomial theorem,

$$(w_jY_0 + w_{j-i}Y_i)^{I_j} = \sum_{m=0}^{I_j} \binom{I_j}{m} w_j^{I_j-m} w_{j-i}^m Y_0^{I_j-m} Y_i^m. \quad (29)$$

**Remark** (Latent-partition interpretation). Equivalently, by Poisson superposition we may regard  $I_j$  as the sum of two independent Poissons — one with mean  $R_0w_jY_0$  counting day- $j$  infections that originated from the day-0 cohort, and one with mean  $R_0w_{j-i}Y_i$  counting those from the day- $i$  cohort. Each term  $m$  in (29) is the joint probability that exactly  $I_j - m$  originate from  $Y_0$  and  $m$  from  $Y_i$ .

**Step 2: joint density.** Multiplying (28) by the three independent priors  $\text{Gamma}(kI_0, k)$ ,  $\text{Gamma}(kI_i, k)$ ,  $\text{Gamma}(kI_j, k)$  and inserting (29),

$$\begin{aligned} p(Y_0, Y_i, Y_j, \text{obs}) &= \frac{(R_0w_i)^{I_i} R_0^{I_j}}{I_i! I_j!} \frac{k^{kI_0 + kI_i + kI_j}}{\Gamma(kI_0) \Gamma(kI_i) \Gamma(kI_j)} Y_j^{kI_j-1} e^{-(k+R_0F_{r-j})Y_j} \\ &\quad \times \sum_{m=0}^{I_j} \binom{I_j}{m} w_j^{I_j-m} w_{j-i}^m Y_0^{kI_0 + I_i + I_j - m - 1} e^{-(k+R_0F_r)Y_0} Y_i^{kI_i + m - 1} e^{-(k+R_0F_{r-i})Y_i}. \end{aligned} \quad (30)$$

This is a constant times a Gamma kernel in  $Y_j$  (the same in every mixture component) times a finite mixture (over  $m \in \{0, \dots, I_j\}$ ) of Gamma-product kernels in  $(Y_0, Y_i)$ .

**Step 3: likelihood.** Integrating (30) via the Gamma normalisation in each variable, define the per-mixture-component constants

$$c_m := \binom{I_j}{m} w_j^{I_j-m} w_{j-i}^m \frac{\Gamma(kI_0 + I_i + I_j - m)}{(k + R_0 F_r)^{kI_0 + I_i + I_j - m}} \frac{\Gamma(kI_i + m)}{(k + R_0 F_{r-i})^{kI_i + m}}. \quad (31)$$

Cancelling the  $\Gamma(kI_j)$  factors and grouping  $k^{kI_j}$  with the  $Y_j$ -rate gives

$$L_{\text{SSI}}^{(\text{iii})} = \frac{(R_0 w_i)^{I_i} R_0^{I_j}}{I_i! I_j!} \frac{k^{kI_0 + kI_i}}{\Gamma(kI_0) \Gamma(kI_i)} \left( \frac{k}{k + R_0 F_{r-j}} \right)^{kI_j} \sum_{m=0}^{I_j} c_m. \quad (32)$$

Setting  $I_j = 0$  collapses the sum to its  $m = 0$  term and the  $(k/(k + R_0 F_{r-j}))^{kI_j}$  factor to 1, recovering (25).

**Step 4: posterior.** Dividing (30) by (32): the  $Y_j$ -Gamma rate-and-shape pair appears in both numerator and denominator and gives the  $\text{Gamma}(Y_j \mid kI_j, k + R_0 F_{r-j})$  density. Within the sum, the constants outside cancel; what remains in the  $m$ th term is  $c_m$  (numerator) divided by  $\sum_{m'} c_{m'}$  (denominator), times the two unnormalised  $(Y_0, Y_i)$  Gamma kernels each with their proper normalising constants supplied directly from the  $c_m$  in the denominator. The result is a finite mixture of independent Gamma products,

$$p(Y_0, Y_i, Y_j \mid \text{obs}) = \text{Gamma}(Y_j \mid kI_j, k + R_0 F_{r-j}) \sum_{m=0}^{I_j} \rho_m \text{Gamma}(Y_0 \mid kI_0 + I_i + I_j - m, k + R_0 F_r) \text{Gamma}(Y_i \mid kI_i + m, k + R_0 F_{r-i}), \quad (33)$$

with normalised mixture weights

$$\rho_m = \frac{c_m}{\sum_{m'=0}^{I_j} c_{m'}}. \quad (34)$$

The mixture index  $m$  tracks how many of the  $I_j$  day- $j$  infections are attributed to  $Y_i$  versus  $Y_0$ ; the weights  $\rho_m$  encode which partition is most consistent with the prior infectivity distributions, the serial-interval weights  $w_j, w_{j-i}$ , and the dispersion  $k$ .

**Step 5: residual extinction probability and PMO.** The remaining-offspring rate is now  $R_0 \Lambda = R_0[Y_0(1 - F_r) + Y_i(1 - F_{r-i}) + Y_j(1 - F_{r-j})]$ , so the conditional extinction probability is  $q_r \mid (Y_0, Y_i, Y_j) = e^{-R_0(1-q)\Lambda}$ . Marginalising over the posterior (33),  $Y_j$ 's expectation factors out (its density is independent of  $m$ ); within each mixture component the  $(Y_0, Y_i)$ -densities are independent Gammas whose MGFs evaluate term-by-term, giving

$$q_r = \beta_j^{-kI_j} \sum_{m=0}^{I_j} \rho_m \beta_0^{-(kI_0 + I_i + I_j - m)} \beta_i^{-(kI_i + m)}, \quad (35)$$

where, for  $a \in \{0, i, j\}$  (with the convention  $F_{r-0} = F_r$ ),

$$\beta_a = 1 + \frac{R_0(1-q)(1 - F_{r-a})}{k + R_0 F_{r-a}}.$$

**Remark** (Reductions to simpler cases). When  $I_j \rightarrow 0$  the sum collapses to its  $m = 0$  term ( $\rho_0 = 1$ ) and  $\beta_j^{-kI_j} \rightarrow 1$ : (35) reduces to (27). When also  $I_i \rightarrow 0$ , the exponent on  $\beta_i$  in (27) vanishes and the exponent on  $\beta_0$  reduces from  $-(kI_0 + I_i)$  to  $-kI_0$ , recovering (20). Similarly (32) reduces to (25) and then to (18).

**Computational note.** The  $c_m$  involve  $\Gamma$  factors of size up to  $kI_0 + I_i + I_j$ , so for moderate  $I_j$  a log-domain computation ( $\log \Gamma$ ,  $\text{logsumexp}$ ) is recommended. The number of sum terms is  $I_j + 1$ , typically small for the histories of interest.